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A comprehensive study on the influence of spatial power distribution on time-dependent keyhole behavior in laser beam welding of copper by means of high-speed synchrotron X-ray imaging☆

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ABSTRACT

This paper examines the impact of spatial power distributions on the time-dependent keyhole behavior during laser beam welding of copper using high-speed synchrotron X-ray imaging. The experimental setup utilized a COHERENT HighLight FL8000-ARM fiber laser with concentric intensity distribution created by an optical fiber cable. The European Synchrotron Radiation Facility (ESRF, beamline ID19) was used to conduct high-speed synchrotron imaging at 20,000 images per second to study the spatio-temporal keyhole behavior. Keyhole geometries were extracted through advanced image processing techniques, allowing quantification of parameters like depth, aperture, bulging, and determination of related oscillation frequencies. The results showed that core-dominated processes exhibit significant variations in keyhole geometry. In contrast, ring-dominated processes exhibited reduced penetration depths but increased melt pool dynamics due to altered absorption conditions and increased temperatures within the melt pool. A stabilized core-ring power distribution minimized fluctuations, resulting in improved process stability. The findings were summarized in a model concept describing three characteristic keyhole regimes: core-dominated, ring-dominated, and stabilized core-ring processes.

1. Introduction

The growing demand for electric vehicles and the associated progress in the field of electromobility have necessitated the development of efficient manufacturing techniques, especially for joining copper. Laser beam welding has gained significant importance for various applications in this field due to its advantages in terms of short processing times, non-contact energy input and high degree of automation. The processing of copper materials is particularly important because of its favorable electrical properties in various components, such as battery technology [1], power electronics or on-board power devices [2] and electric motors [3]. In order to achieve or maintain the required electrical and mechanical properties after welding, it is essential to implement low-defect processes. This typically involves the consideration of spatters and pores, which reduce the current-carrying cross-sectional area of the conductor due to loss of material or voids [3]. In addition, typical electric vehicle applications involve a large number of welds in each component, e.g., up to 220 hairpins per stator [4], setting high demands on process capability [5].

High power near-infrared wavelengths of high brilliance are often used for welding copper due to the wide use of solid-state lasers, i.e., fiber lasers [6] or disk lasers [3]. This is accompanied by a low

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absorption coefficient in the solid state of copper and an instantaneous increase in the transition to the melt [7], as well as a simultaneous reduction in the thermal conductivity by half [8], which supports an abrupt vaporization of the material and the formation of a keyhole. The keyhole formation depends on the pressure balance (Equation (1)) between opening and closing pressures, considering the recoil pressure p_{abl}, the differential vapor pressure $\Delta(p_v - p_0)$ with the atmospheric pressure p_0, the surface tension pressure p_σ, the hydrostatic pressure p_h and the hydrodynamic pressure p_{dyn} [9].

$$\underbrace{p_{abl} + \Delta(p_v - p_0)}_{opening\ pressures} = \underbrace{p_\sigma + p_h + p_{dyn}}_{closing\ pressures} \quad (1)$$

The pressure balance is affected by various factors including energy input, intensity distribution, metal vapor, melt flow and atmospheric pressure [10]. The complex interaction between the pressure balance terms and the associated mass, momentum and energy transfer between the keyhole and the surrounding melt pool leads to fluctuations in the laser welding process, especially depending on the welding velocity. These fluctuations of the keyhole depend on the spatial direction [11] and can be described as anharmonic oscillations, e.g., for penetration depth and keyhole aperture [12], and result in non-uniform penetration depths, accelerated melt pool movement around the keyhole causing spatter or pinch-off of the keyhole leading to pore formation [13].

In recent years, progress has been made in understanding the formation of keyhole and melt pool related defects such as spatter [14,15] and pores [13]. It is known that as the processing speed increases, the inclination of the keyhole front wall changes, causing differences in the absorption of laser radiation at the keyhole front [16]. Thereby, laser radiation is mostly absorbed at the keyhole front wall, however, the resulting laser intensity distribution on front and rear wall also depends on multiple reflections within the keyhole [17]. A quasi-steady-state model with a cylindrical heat source, considering both convection and conduction, enabled plotting the absorbed laser intensity and dissipated heat flux against the depth of the keyhole. It can be concluded that a lower amount of laser radiation is absorbed at the rear wall compared to the front wall, as the melt flows from the front to the rear side of the keyhole. The absorption at the front wall is also linked to the formation of humps as described and discussed in [14]. The resulting momentum transfer from local evaporation leads to an increase in velocity at the melting layer between keyhole front and solid base material. This significantly affects the flow field and supports the downward flow of molten material to the bottom of the keyhole and consequently the upward flow of melt at the back of the keyhole. The interaction between the metal vapor and the upwardly moving melt at the rear wall leads to fluctuations of the keyhole. It has been noted that necking can occur at the top of the keyhole. In the subsequent time period, the melt flow is accelerated, resulting in widening the keyhole and forming a melt column that rises above the melt pool. Depending on the kinetic energy of the fluid, molten material may detach in the form of spatters [18]. In addition, melt ejections from bulge formation are well known for welding copper [19]. Fluctuations in the keyhole bottom can change the angle of incidence and result in varying absorption conditions. Increased local evaporation can cause a bulge due to the increased opening pressures acting in this region. The bulge forces the melt towards the sample surface, resulting in melt ejection when the surface tension of the melt is exceeded. Furthermore, fluctuations in evaporation processes lead to significant dynamics in the keyhole and melt pool, which can cause instabilities and defects such as pores [13]. In addition, at high vapor velocities, friction between metal vapor and keyhole wall can result in the detachment of small, undirected spatters [20].

In order to avoid such defects, different empirical approaches were developed in recent years for near infrared laser processes. For instance, power modulation [21], welding under reduced ambient pressure [22,23] and utilization of beam shaping have shown a high potential [13,24]. Beam shaping in terms of concentric intensity distributions realized by optical fiber cables [25] are of specific interest as commercial systems such as COHERENT Adjustable Ring Mode (ARM), IPG Adjustable Mode Beam (AMB) or Trumpf BrightLineWeld are available. It is important to acknowledge that the architecture of the beam source can vary depending on the manufacturer, yet the fundamental principle of the core-ring fibers remains comparable. Kliner et al. provided a comparison between fiber cable-based beam shaping to alternative beam-shaping techniques [26].

Fig. 1 provides a schematic illustration of laser deep penetration welding with concentric intensity distributions, when core and ring intensities contribute to the formation of the keyhole. In general, the utilization of spatial beam shaping such as concentric intensities allows to manipulate the temperature distribution in and around the keyhole and therefore acts actively on the process, as shown for steel [27,28]. The behavior of the melt is affected by the temperature as well as by the surrounding atmosphere due to surface tension [29] and interaction with ambient gases [30]. On the one hand, the surface tension of molten copper is reduced by increasing temperature [31], i.e., the surface tension pressure term p_σ is reduced. On the other hand, different gaseous elements can lead to porosity depending on the copper base material [30]. Florian et al. demonstrated that for concentric intensities, core-dominated processes tend to generate irregularities such as spatter, melt ejection, and pores originating from the interaction between keyhole and melt pool [13]. A balanced power distribution between core and ring intensities prevented porosity and produced a uniform weld appearance, while ring-dominated processes resulted in weld surface irregularities.

However, the effect of such intensity profiles on the keyhole shape, on the intensity distribution within the keyhole as well as on the dynamic behavior during copper welding is not yet described in detail. The highly dynamic phenomena driven by the rapid interaction between keyhole and melt pool are responsible for resulting defects [12,13]. Their description requires advanced characterization techniques and the ability to reveal phenomena hidden in the material. For this reason, *in situ* X-ray imaging is of high interest. Most studies from the past were able to address acquisition rates of 1,000 [32] up to 5,000 images per second [33] and provided insights into keyhole geometries for different welding speeds and wavelengths [32,33]. However, the utilization of polychromatic synchrotron radiation has been shown to enhance the temporal and spatial resolution of radiography coupled with more sensitive contrast modes [34]. This has enabled acquisition rates up to 20,000 images/second with excellent signal to noise ratio during laser beam welding of copper, providing valuable insights into the time-dependent behavior of keyhole and melt pool phenomena [12,13]. Using such capabilities provide a high spatial and temporal resolution of processes hidden in the workpiece to describe the influence of the

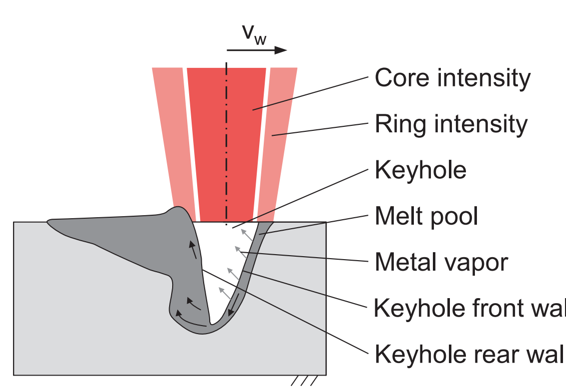

Fig. 1. Schematic representation on laser welding with concentric intensity distributions. Depending on the ratio between the keyhole aperture and the diameter of the ring, laser beam can irradiate the melt film on the front as well as the melt on the rear wall of the keyhole, influencing the temperature distribution.

concentric intensity distributions.

In this paper, an *in situ* high-speed synchrotron imaging laser welding setup was used to systematically study the effects of varied concentric power distributions between core and ring during welding of copper at 20,000 images/second. Based on image processing and extraction of quantified measures, an in-depth analysis of time-dependent effects of keyhole and melt pool, provided novel insights for increasing the fundamental understanding of copper welding processes with different intensity profiles. The gradual adjustment of power distribution between the core beam and the ring beam alters the weld depth, absorption, and pressure balance. This, in turn, results in varying interactions between the keyhole and the melt pool and allows for systematic identification and description of the relevant effects and influencing factors.

2. Materials and methods

2.1. Experimental setup

Fig. 2a shows the setup for laser welding and high-speed synchrotron X-ray imaging. A COHERENT HighLight FL8000-ARM fiber laser was used for carrying out the welding experiments in combination with a Precitec YW52 processing head (incidence angle: 0°). The laser beam source provided a laser focal geometry, consisting of a ring and a core aligned concentrically with a wavelength of 1,070 nm. The magnification ratio of fiber configuration and processing head led to a core diameter of 89 µm, an inner ring diameter of 109 µm and an outer ring diameter of 208 µm. Fig. 2b provides a comparison between core and ring beam profile and intensity (D86 method, measured at 0.3 kW with Primes FocusMonitor to stay below the damaging threshold of the device), d_f indicates the individual beam diameter in the focal plane. The beam source allowed the adjustment of laser beam power P_L in core and ring up to 4 kW each. Preliminary trials ensured the penetration depth of the weld remained within the region of interest of the high-speed synchrotron imaging setup, which is why a maximum beam power of 3.5 kW was chosen for a constant welding speed ($v_W = 10 \frac{m}{min}$). In order to systematically study the effect of beam power distribution, the power was varied between core (P_{core}) and ring (P_{ring}) in steps of 0.5 kW from 0 kW to 3.5 kW (Table 1). The focal plane was set at the surface of the copper specimen. The setup used is identical to [12] and [13].

Copper Cu–ETP (CW004A, 2.0065) was chosen as material. The movement v_M of the sample was realized by a linear axis while v_W represents the resulting welding direction. This procedure ensured that the region of interest of the radiography, i.e., the interaction zone between laser beam and material, stayed in the same place over all experiments and allows for comparing the resulting keyhole geometry. It should be noted that the sample was accelerated to the welding speed before the welding process was started.

Table 1
Spatial power distribution used for different steps of core power P_{core}, ring power P_{ring} and total beam power..P_L

Power in core beam P_{core} / kW	Power in ring beam P_{ring} / kW	Toal beam power $P_L = P_{core} + P_{ring}$ / kW
3.5	0	3.5
3.0	0.5	3.5
2.5	1.0	3.5
2.0	1.5	3.5
1.5	2.0	3.5
1.0	2.5	3.5
0.5	3.0	3.5
0	3.5	3.5

The sample width was chosen to be 2 mm to ensure a sufficient frame rate in high-speed synchrotron X-ray imaging and to minimize the effect of heat accumulation during the welding process under different intensities. The high-speed synchrotron X-ray imaging was carried out at beamline ID19 of the European Synchrotron Radiation Facility (ESRF) with an X-ray energy of ≤ 60 keV and a beam size of 4 x 4 mm^2. A Ce-doped $Lu_3Al_5O_{12}$ scintillator with a thickness of 250 µm was used. A sample to scintillator distance of 4.5 m was achieved to benefit from the propagation-based phase contrast. The scintillator was lens-coupled (5x/0.14 NA) to a high-speed camera Photron SA–Z obtaining 20,000 images/second with a sensor resolution of 1024 x 1024 $pixel^2$ at 4 µm edge length per pixel.

2.2. Image processing and data evaluation

A modified flat field correction was carried out for pre-processing the high-speed recordings. Therefore, a median image $\bar{I}$ consisting of 500 frames was created with the sample moving at the welding speed but without carrying out the laser welding process. This reference $\bar{I}$ was

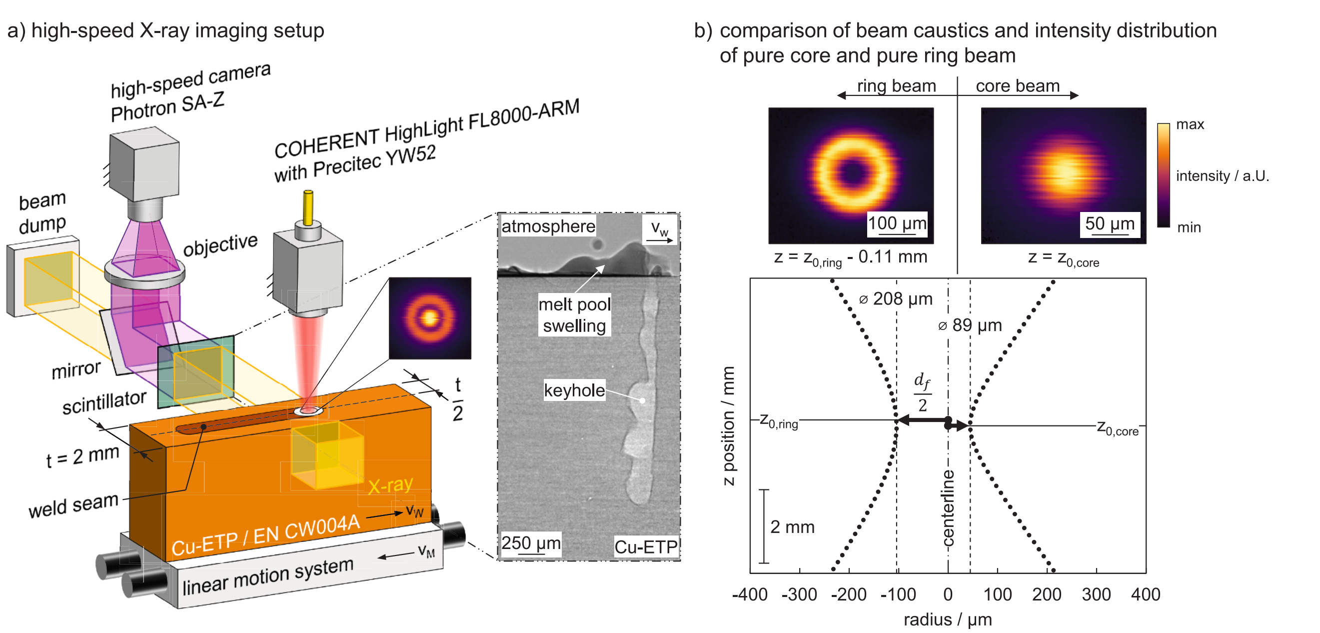

Fig. 2. a) setup for laser welding and high-speed synchrotron x-ray imaging, b) comparison of beam caustics based on the half-side beam profile of core and ring beam (d86 method) and related false color image of intensity distribution.

calculated individually for every video recorded to minimize image processing defects. An individual index n was assigned to each frame I_n of the video during the welding process to ensure that the exact time step and parameter set is known for each image. The modified flat field correction F_n was calculated for each frame individually in accordance with Equation (2). Minor adjustments in brightness and contrast for figures shown in the paper were conducted to maintain a consistent visual appearance.

$$F_n = \frac{I_n}{\bar{I}} \tag{2}$$

Based on the resulting frame F_n, a further image and data processing sequence was carried out in order to individually determine the keyhole geometry for every frame (Fig. 3). A pixel classification approach was realized based on the software ilastik [35] (version 1.4.0) to handle the effects of dynamic keyhole movement, and the effects of high density of copper that result in varying gray values of the keyhole contour. Intensity, edge, and texture were considered as criteria for pixel classification. Up to 30 frames were used as training data for an individual parameter set. The pixel classification resulted in a segmentation between base material and gaseous phases including the keyhole and pores. This procedure allowed the binarization of the image and further image processing setups using MATLAB 2022a. These additional processing setups mainly addressed the removal of artefacts, e.g., pores. The image processing steps enabled the keyhole geometry to be determined for each individual frame (G_n) and to calculate geometrical parameters, i.e., the keyhole aperture at the surface d, the keyhole depth t_d, the greatest bulging characterized by maximum keyhole length b and the depth of the maximum keyhole length t_b (Fig. 3). This evaluation allows the quantification of keyhole geometries and comparison between different power levels in core and ring. The position of the maximum bulging relative to the keyhole depth [12], defined as TB_{rel}, compares the different intensities independently from the depth of penetration, and can be calculated according to Equation (3).

$$TB_{rel} = \frac{t_b}{t_d} \tag{3}$$

1,000 consecutive frames from the weld seam center were evaluated for each parameter to ensure that a sufficient sample size was considered. It should be noted that major deviations, for instance, originating from defects like large melt ejections, that were affecting the characteristic keyhole regime, were excluded from the evaluation. The evaluation of a series of consecutive frames allowed further data analysis using Fast Fourier Transform (FFT) to determine the oscillations of keyhole aperture d and keyhole depth t_d.

A time-dependent representation of individually binarized keyhole geometries (G_n) enables further understanding of the temporal behavior of different effects. For example, the movement of bulges in time steps of 0.05 ms based on an acquisition rate of 20,000 Hz in high-speed X-ray imaging. On the other hand, further information on characteristic keyhole regimes can be provided by utilizing sum images. A sum image G_{sum} was calculated based on arithmetic addition of 500 individually binarized keyhole geometries G_n for each power distribution to provide a further understanding of the keyhole geometry, and its deviation (Equation (4)). Using a false-color representation, it was possible to show where keyhole geometry occurs more or less frequently.

$$G_{sum} = \sum_{n=1}^{500} G_n \tag{4}$$

2.3. Ray tracing model

The 500 binarized keyhole geometries also allowed calculation of median keyhole geometries. These median geometries, depending on different power distributions between core and ring, were used to calculate the absorbed intensity along the keyhole wall. This was realized by a simplified 2D ray tracing model based on Comsol Multiphysics 6.1 (Ray Optics Module, documentation in [36]), considering the propagation of the laser beam within the keyhole and its specular reflection at the keyhole wall (Fig. 4). The absorptivity A_i for each reflection at the keyhole wall was considered constant at 14.5 % as the maximum value at 1400 K based on [33]. A free triangular mesh was used. Element sizes between 0.2 and 5 μm provided a sufficient representation of the keyhole geometry as well as for core and ring beam. It should be noted that temperature-dependent effects, the interaction between the laser beam and (condensed) metal vapor, or heat transfer were not considered.

The intensity distributions of core and ring beam were assumed to be top hat distributions, i.e., the beam power was uniformly distributed

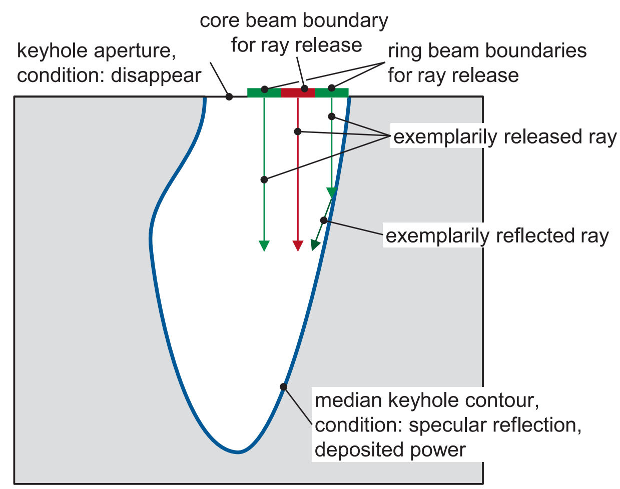

Fig. 4. Schematic representation of the ray tracing model with boundary conditions.

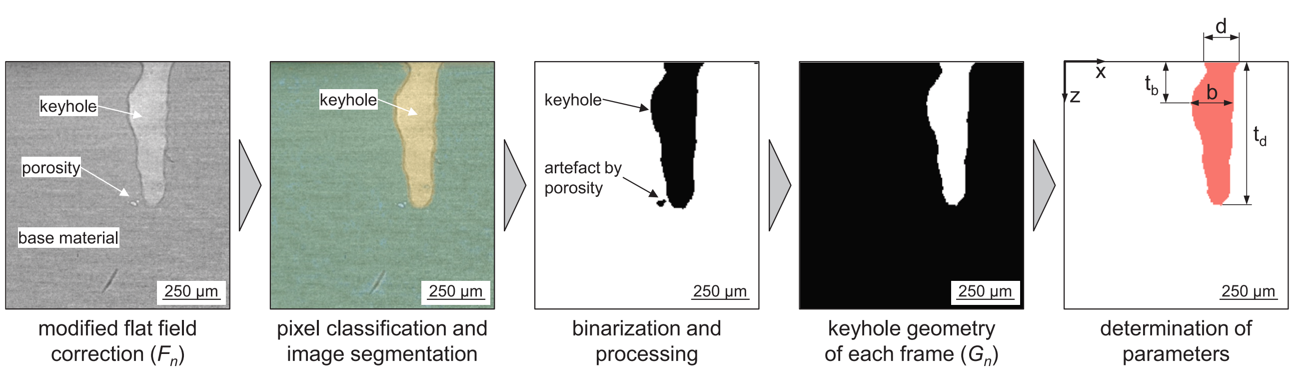

Fig. 3. Image and data processing on the example of an individual frame F_n.

among the rays. The geometry of core and ring beam in the focal plane were implemented as boundaries with the dimensions determined in Section 2.1. The boundary was located at the keyhole top surface comparable to the focal position in the experiment. Individual divergence angles α were considered to match the real laser beam caustic in the model. In order to provide a sufficient representation of the core and ring beam, 50,000 rays were employed for each uniformly distributed over the boundary. The rays were released as unpolarized rays at the boundaries. The initialization of the ray release was equivalent to the first time step in the calculation. The ray trajectory was calculated based on a system of coupled Hamilton equations, considering the initial ray coordinates $\overrightarrow{q}$, wave vectors $\overrightarrow{k}$ and the angular frequency ω, where $n(\overrightarrow{q})$ is the refractive index and c is the speed of light in vacuum (Equation (5)) [36].

$$\frac{d\overrightarrow{k}}{dt} = -\frac{\partial\omega}{\partial\overrightarrow{q}}; \frac{d\overrightarrow{q}}{dt} = \frac{\partial\omega}{\partial\overrightarrow{k}}; \omega = \frac{c\left|\overrightarrow{k}\right|}{n(\overrightarrow{q})} \tag{5}$$

The trajectory of each individual ray was calculated on this basis. The keyhole geometry was modelled as a wall with specular reflection so that the angle of incidence and the angle of reflection are equal. When a ray reached the keyhole wall during propagation, the angle of incidence Θ_i was computed following Snell's law, where n_i is the unit vector in the direction of the incident ray and n_s is a unit vector normal to the boundary (Equation (6)). On this basis, the reflected ray n_r at the boundary was calculated (Equation (7)) [36].

$$\Theta_i = \arccos\left(\frac{\overrightarrow{n_i}\overrightarrow{n_s}}{|\overrightarrow{n_i}||\overrightarrow{n_s}|}\right) \tag{6}$$

$$\overrightarrow{n_r} = \overrightarrow{n_i} - 2\overrightarrow{n_s}cos\Theta_i \tag{7}$$

The fluence rate E_0 representing the specific energy flux on the keyhole wall was calculated (Equation (8)) where Q_i represented the individual ray power, c_i the speed of light of the ray, and V_m the mesh element volume. It should be noted that E_0 was accumulated for each mesh element, i.e., the exact position of the ray within a single element was not considered, but the small mesh sizes provided a sufficiently high spatial resolution. This provided a value for the absorbed intensity I_{abs} at different positions considering multiple reflections as accumulated variable upon reaching the full calculation time [36].

$$\frac{\partial E_0}{\partial t} = \frac{1}{V_m}\sum Q_i c_i \tag{8}$$

The calculation was carried out over a duration of 50 picoseconds with timesteps of 0.1 picoseconds to ensure that no further significant increase of absorbed intensity was noticeable at the keyhole wall. Individual rays that reached the keyhole aperture or the boundaries of core and ring beam due to reflections, disappeared due to an additional boundary condition that eliminates parasitic reflections [36].

3. Results and discussion

3.1. General process behavior exemplarily described based on high-speed radiography

First, an overview of the deep penetration welding process of copper depending on the spatial power distribution will be provided. The laser beam power used for the welding process was kept constant at 3.5 kW. The power was varied between core P_{core} and ring P_{ring} in steps of 0.5 kW from 0 kW to 3.5 kW (Section 2.1) to systematically study the effect of spatial power distribution. This approach enabled the development of a fundamental understanding of spatial power distribution and its impact on keyhole behavior. It is important to note that this approach differs from strictly application-oriented studies, which typically operate at constant welding depths but with adapted total power.

Fig. 5 presents a characteristic image series with minimal time steps of 0.05 ms between the individual images to provide an insight into the time-dependent process behavior. First conclusions about the dynamics of the process can be made on the basis of the high temporal resolution. The respective parameters are indicated accordingly and show the process with full core power ($P_{core} = 3.5\ kW$, Fig. 5a), with core-ring power distribution ($P_{core} = 1.5\ kW$, $P_{ring} = 2.0\ kW$, Fig. 5b), and with full ring power ($P_{ring} = 3.5\ kW$, Fig. 5c). The same scale between Fig. 5a-c allows for a direct comparison of different parameters.

Welding with full core power ($P_{core} = 3.5\ kW$, Fig. 5a and Video 1) is characterized by strong fluctuations between the individual timesteps. The first image, representing $t = t_0$, shows a slender keyhole geometry with high aspect ratio between keyhole aperture and keyhole depth without significant bulges. The following image ($t = t_0 + 0.05\ ms$) shows strong shape deviations at the front and rear wall of the keyhole, which illustrates the rapid changes within the keyhole. This behavior is related to significant fluctuations in the spatio-temporal absorption conditions, as the angle of incidence of the laser beam can vary widely, which becomes apparent when directly comparing different time steps. This can also be seen in the following image ($t = t_0 + 0.20\cdots0.25\ ms$), as bulges appear at the bottom of the keyhole (bulge I) and in the center of the keyhole (bulge II). The geometry of the keyhole resembles the 'J-shaped' keyhole observed for laser material processing of other materials, such as aluminum [37] and titanium [38]. The bulges vary in size or new bulges occur at other locations (bulge III, $= t_0 + 0.30\ ms$). These bulges can follow a highly dynamic behavior, for example, by fast deformations induced by absorption peaks or by the metal vapor flow ($t = t_0 + 0.30\cdots0.40\ ms$). It should be noted that the greatest expansion of the keyhole typically occurs inside the workpiece and is not characterized by the keyhole opening at the top surface.

A change in the power distribution has a significant effect on the appearance of the keyhole as represented by Fig. 5b for the selected core-ring intensity ($P_{core} = 1.5\ kW$, $P_{ring} = 2.0\ kW$, Video 2). Most noticeable is the major change in aspect ratio due to the larger beam diameter at the top of the keyhole and the reduced depth due to the power shift from core to ring intensity. The radius at the keyhole leading edge in welding direction is much more pronounced than with full core power welding ($t = t_0$). Fluctuations also occur at the front and back of the keyhole, but with reduced dynamics, while the leading edge of the keyhole in welding direction is significantly less fluctuating ($t = t_0 + 0.05\cdots0.30\ ms$). It has been observed that the bulges occur primarily in the upper area of the keyhole (bulge I), while the lower area is subject to less fluctuation (bulge II).

A further change in the aspect ratio can be seen with greatly reduced keyhole depth ($t = t_0$) when using the pure ring intensity (Fig. 5c). The inclination angle of the keyhole front is significantly declined and less steep due to the changed absorption conditions between the core and ring intensities ($t = t_0 + 0.05\ ms$). In addition, the height of the melt pool wave in the region of the rear wall of the keyhole is significantly increased compared to Fig. 3a-b. The keyhole region demands more space relative to the weld pool volume, resulting in greater material displacement. Considering the selected time interval ($t = t_0 + 0.05\cdots0.30\ ms$), the keyhole seems to be less subject to fluctuations; shape deviations are mainly recognizable by small spikes in the rear wall. The momentum transfer between metal vapor and melt pool wave at the rear wall is increased due to the reduced keyhole front angle, resulting in a larger keyhole aperture and intensified melt pool dynamics ($t = t_0 + 0.15\ ms$). It should be noted that an increased occurrence of pore formation can be observed that is not linked to instabilities of the keyhole (i.a. at $= t_0 + 0.15\ ms$). The emergence of new pores is observed in the upper area of the keyhole and melt pool waves facing the keyhole in this region (Video 3). Additionally, the changes in the escaping metal vapor perpendicular to the surface can be associated with an increased interaction between metal vapor and the melt pool,

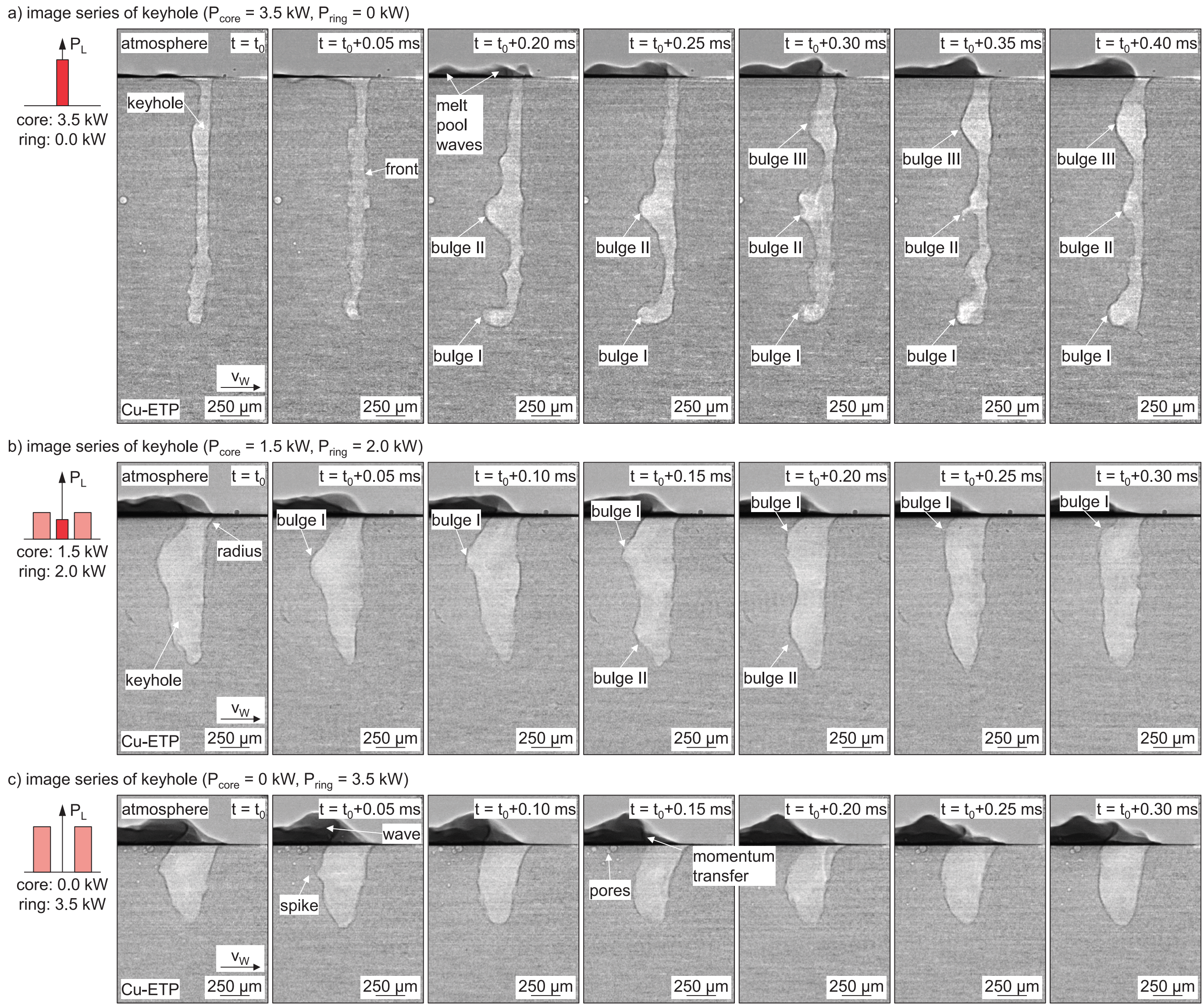

Fig. 5. Exemplary radiography image series for a) P_{core} = 3.5 kW, b) P_{core} = 1.5 kW / P_{ring} = 2.0 kW and c) P_{ring} = 3.5 kW.

leading to an increased keyhole aperture at the surface as mentioned before. Consequently, surface temperatures of the melt pool in this region are expected to be higher compared to slender keyhole regimes with steep inclination angles at the front. In addition, the altered absorption conditions at the keyhole rear wall can contribute to an increase in temperature (Section 3.4). These assumptions are supported by the simulation presented in [13] in the context of the melt pool temperature distribution. Alter et al. assumed that increased nitrogen dissociation can be a precondition for increased pore formation in Cu-ETP [30]. The elevated temperatures in this region may facilitate the dissociation of nitrogen from the atmosphere. In addition, the increased size of the keyhole aperture leads to an enlarged region of interaction with the atmosphere — conditions that are absent in other keyhole regimes with slender geometry and steep front angle. Section 3.4 provides further insights based on spatially resolved calculation of absorbed laser beam power by means of ray tracing.

3.2. Evaluation of time-dependent keyhole behavior

Section 3.1 provided an example of how the keyhole behaves depending on different power distributions. The following section provides a description and illustration of variations in keyhole geometry over an extended period. To present a longer observation period compactly for each recorded time step during radiography, keyhole geometries were extracted from all images recorded using the image and data processing pipeline, as explained in Section 2. The extracted geometries were sequenced from left to right in order of increasing time and shown as a false-color plot to enable the temporal assignment of each geometry. Consequently, each keyhole geometry represents an individual frame recorded at 0.05 *ms* intervals. The sequence of images starts at $t = t_d$ (shown in blue) and ends at $t = 2$ *ms* (shown in yellow). Note that different scales are used in the weld and in the depth directions to compare the different power distributions. Fig. 6 shows a series of keyhole geometries for different power distributions, as described, to provide further information on the characteristic behavior.

Starting with full core power, the behavior is shown (Fig. 6a). The keyhole sequence shows the slender appearance of the keyhole and its rapid changes in weld depth, indicating spiking at the keyhole bottom (approx. $t = 1.1\cdots1.2$ *ms*) and the behavior of bulges, including their growth and fusion (approx. $t = 1.35\cdots1.6$ *ms*). When combining ring and core power ($P_{core} = 3.0\ kW$, $P_{ring} = 0.5\ kW$ Fig. 6b), the frequency of occurrence and the size of the bulges decreases over the considered time interval, although keyhole depth is still fluctuating. As the power of the ring increases ($P_{core} = 2.0\ kW$, $P_{ring} = 1.5\ kW$ Fig. 6c), a distinct

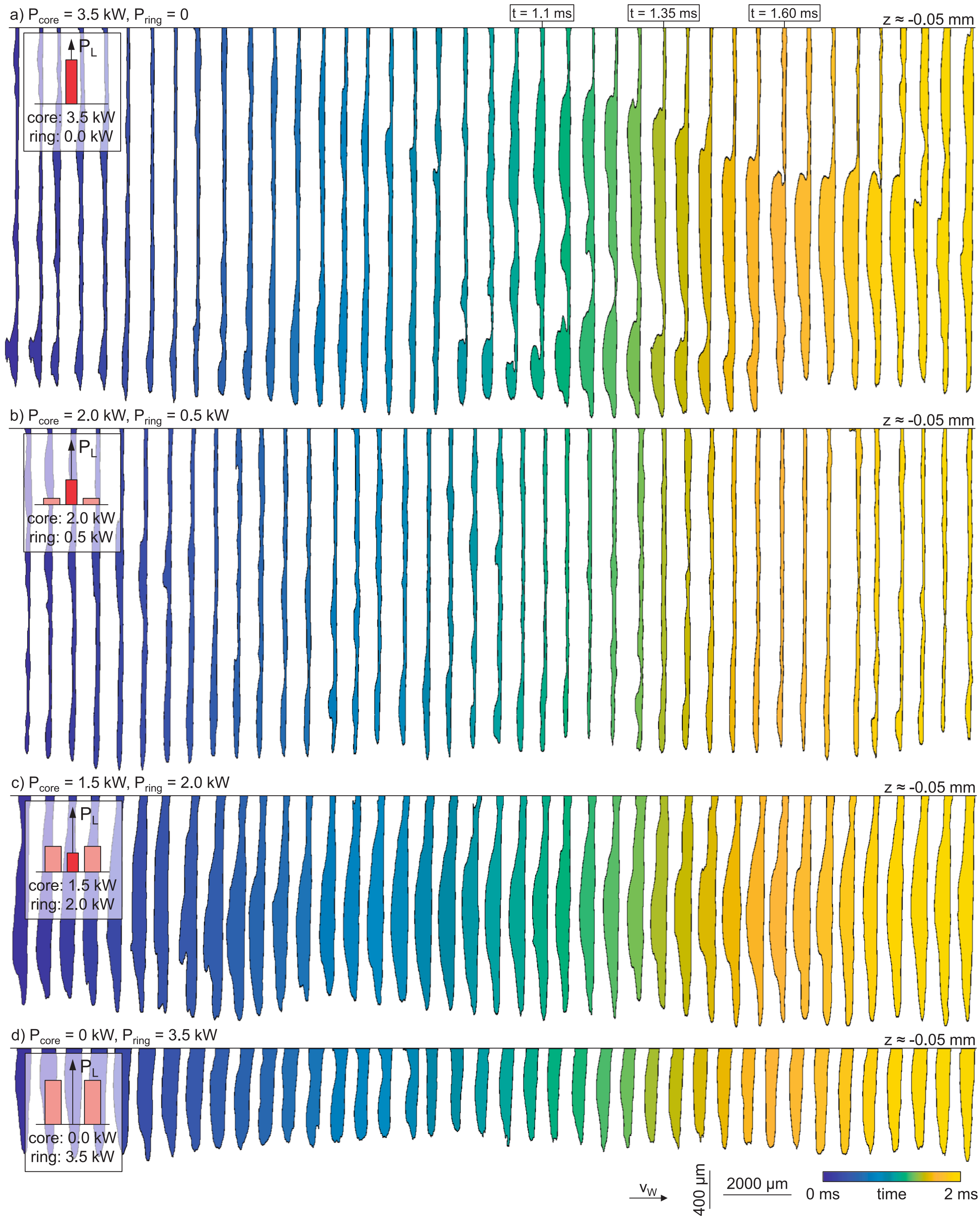

Fig. 6. False-color plot for tracking of keyhole geometry over time for different power distributions: a) $P_{core} = 3.5$ kW, $P_{ring} = 0$ kW, b) $P_{core} = 3.0$ kW, $P_{ring} = 0.5$ kW, c) $P_{core} = 1.5$ kW, $P_{ring} = 2.0$ kW, d) $P_{core} = 0$ kW,.$P_{ring} = 3.5$ kW

bulge in the center of the keyhole can be seen with less fluctuations in welding direction. The keyhole depth is reduced and appears to fluctuate less rapidly than at high core power. Due to the high temporal resolution and the inertia of the melt due to heat conduction, this effect may be insignificant in the resulting weld depth. It appears that the keyhole depth fluctuates at a reduced frequency when welding at full ring power ($P_{ring} = 3.5\ kW$, Fig. 6d), with an even greater reduction in the aspect ratio of the keyhole aperture to the weld depth. In nearly all images considered for the depicted time series, the keyhole aperture at the top is smaller than the inner extent of the keyhole below the surface.

In summary, the analysis of the keyhole sequence over time allows for a more comprehensive analysis of the keyhole behavior and further quantification of changes. On this basis, the extracted keyhole geometries can be used to evaluate bulging behavior, keyhole width and depth, and to calculate oscillation frequencies. 1,000 individual images for each power distribution were considered to provide a large database for describing changes in keyhole geometry during the laser welding process. It should be noted that large defects such as the ejection of the whole melt pool were not appearing within the 1,000 frames chosen.

3.3. Quantification of geometrical parameters

Different geometrical parameters were quantified based on the image processing method described in Section 2.2. The bulging explains

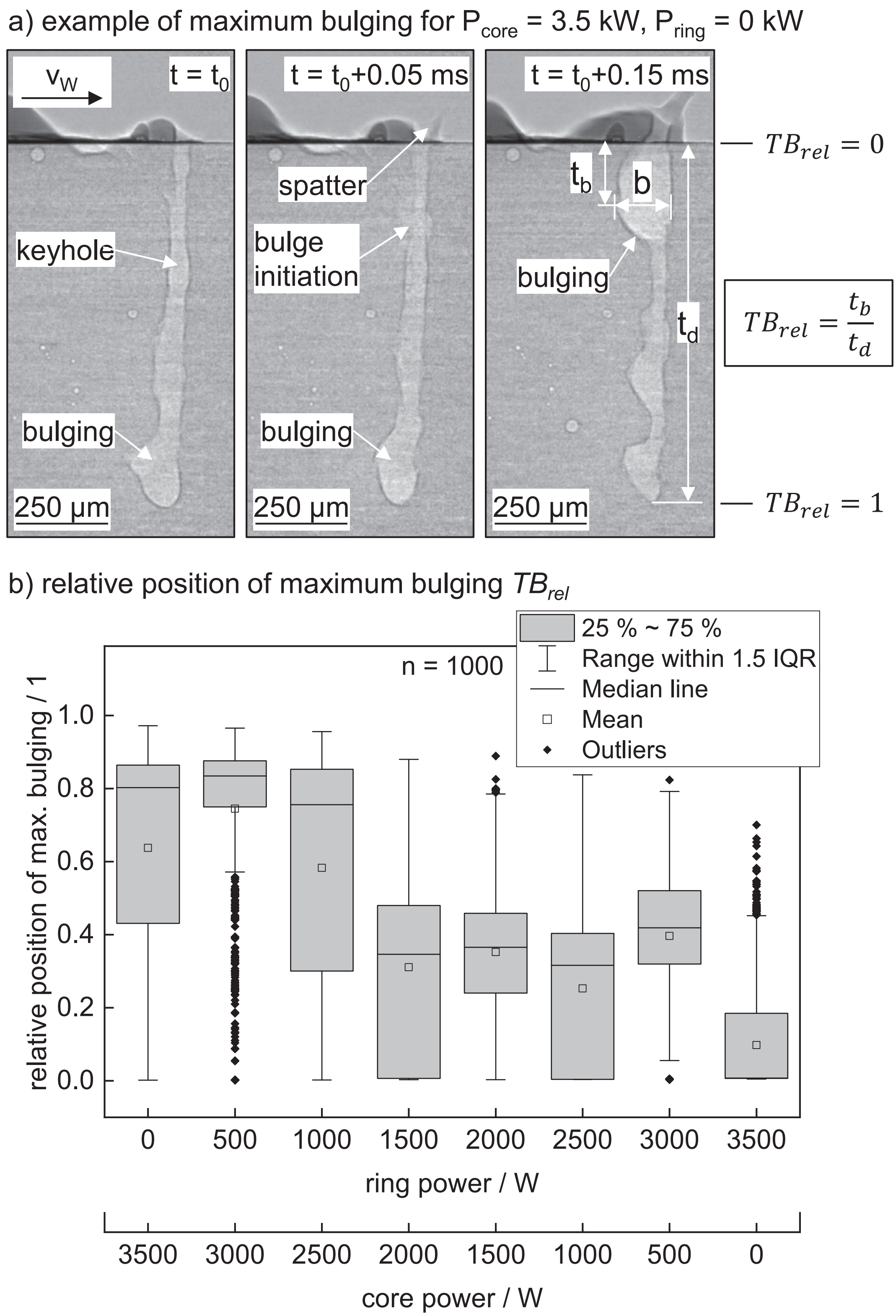

Fig. 7. a) example of maximum bulging at $P_{core} = 3.5$ kW, $P_{ring} = 0$kW, b) relative position of maximum bulging (TB_{rel}) for different ratios of core and ring power.

the present deformation of the keyhole contour and will be evaluated in the first step, as it is highly relevant to occur during pore formation [13,39,40].

Laser beam welding with different power distributions is associated with different keyhole depths, and hence, a basis for comparing different keyhole regimes is necessary. Thus, the location of maximum bulge width in welding direction was normalized to the keyhole depth in order to provide the value of TB_{rel} as the basis for comparison. In this case, $TB_{rel} = 1$ corresponds to the bottom of the keyhole and $TB_{rel} = 0$ to the surface of the weld (Section 2.3). Fig. 7a shows an example of the maximum bulging at pure core power intensity. In the image, at $t = t_0$, a minor bulge occurs at the keyhole bottom region. The size of the bulge slightly changes at $= t_0 + 0.05\ ms$, i.e., this bulge would determine TB_{rel}. The initiation of a bulge in the upper keyhole region can be seen due to fluctuations in absorption and increased evaporation in this region. High velocity of vapor escaping the keyhole also may lead to small spatters forming at the keyhole wall. A further increase in time $t = t_0 + 0.15\ ms$ shows that multiple bulges occur at different positions. Thus, the maximum keyhole length in the near-surface region would provide the result for TB_{rel} in this frame. A detailed explanation of TB_{rel} is provided in the context of Equation (3) (Section 2.2). It should be noted that this method neglects the occurrence of multiple bulges at different positions and only considers the maximum keyhole length, while providing the possibility to compare the different power distributions based on the maximum keyhole size.

Fig. 7b shows the relative position of maximum bulging TB_{rel} for all distributions between core and ring power as box plot, to provide a nuanced view considering the mean value (□), the median (–), the 1.5 inter quartiles range (IQR, whisker), the outliers (◆), as well as the 25 % and 75 % percentile (grey box).

Starting at full power in the core beam ($P_{core} = 3.5\ kW$, $P_{ring} = 0\ kW$), the maximum bulging described by TB_{rel} occurs for the interquartile rang between approx. 0.45 to 0.85, i.e., in the lower half of the keyhole region. This may be explained by absorption peaks in this region and strong deformations of the keyhole front, leading to changing absorption conditions that are related to high vapor velocities and high momentum transfer between vapor and keyhole boundaries. A more detailed look at the spatial distribution of the absorbed laser beam is provided in Section 3.4. Increasing the ring power leads to a stronger focus of the bulges to the keyhole bottom region ($P_{core} = 3.0\ kW$, $P_{ring} = 0.5\ kW$). It can be assumed that the less pronounced bulges, as seen in Section 3.2 and Fig. 6b, indicate reduced fluctuations, which is why bulges occur more consistently in the bottom region. With a further increase of ring power, the median value significantly drops from approx. 0.75 to 0.35 accompanied by a shift of the 25 ∼ 75 % percentile to the upper half of the keyhole ($P_{core} = 2.0\ kW$, $P_{ring} = 1.5\ kW$). The maximum bulging occurs in the upper keyhole region from this value onwards and reaches the most surface-near occurrence of bulges for the full power in the ring ($P_{core} = 0$, $P_{ring} = 3.5\ kW$). The behavior confirms the observations made in Section 3.1 regarding the interaction of the metal vapor with the melt pool wave for this regime. It should be noted that the penetration depth and the aspect ratio of the keyhole are changing depending on the power ratio between core and ring beam.

In addition to this general statement, a large number of outliers in power distributions with core powers of $P_{ring} = 0.5kW$ and $P_{ring} = 3.5kW$ are noticeable. For these two parameters, the interquartile range is very small, meaning a large number of data points are closely grouped. For $P_{ring} = 0.5kW$ ($P_{core} = 3.0kW$), TB_{rel} is close to the bottom of the keyhole, as reflected by the median value, and outliers occur for positions further up in the keyhole. The slim shape of the keyhole and the 'J-shaped' keyhole (Section 3.1) maximize expansion in the lower region, i.e., the bulges in the upper region must exceed this size to be considered as maximum bulging. Therefore, fluctuations in the upper range due to further bulges are less decisive than the effect at the bottom of the keyhole. For $P_{ring} = 3.5kW$ ($P_{core} = 0kW$), the interquartile range of TB_{rel} is very small, ranging between approx. 0.2 and 0 (i.e. close to the surface). Outliers reveal fluctuations that indicate the location of the maximum bulging deeper below the surface. These fluctuations may be caused by keyhole-closing flows at the keyhole aperture, which may be triggered by the melt pool wave in the rear area of the keyhole, as noticeable at the beginning of Video 3.

Fig. 8a provides a further insight into bulging and keyhole geometry based on sum images for selected power distributions. A false-color representation was used to illustrate the characteristic keyhole region and its spatial fluctuations, which were based on the arithmetic addition of individually binarized keyhole geometries (see Section 2.2 and Equation (4)). The core power is declining from the left to the right, the ring power increases vice versa. The maximum core power ($P_{core} = 3.5\ kW$, $P_{ring} = 0kW$) leads to the strongest deviations in the bottom (I) and center region (II). The high penetration depth of the laser beam is accompanied by high hydrostatic pressures, trying to close the keyhole, while changing the absorption conditions occurring at the keyhole front, such as by waves, and thus altering angles of incidence. This leads to transient evaporation processes, dynamically counteracting the closing pressures. If this happens in the upper keyhole region, the high momentum transfer can lead to an increased keyhole opening at the top side. In addition, the upward movement of bulges support deviations in this region. This phenomenon and melt pool waves, oriented contrary to the welding direction, are leading to fluctuations in the upper region of the keyhole rear wall (III). An increase in ring power ($P_{core} = 3.0\ kW$, $P_{ring} = 0.5\ kW$) confirms the more concentrated appearance of the bulges in the keyhole bottom region (IV) and less pronounced fluctuations. A transition mode in the keyhole geometry is observed with a further increase in ring power ($P_{core} = 2.0kW$, $P_{ring} = 1.5kW$). The keyhole fluctuates between a smaller (V) and a significantly larger aspect ratio (VI), i.e., the welding depth and the front angle of the keyhole also change (Video 4). This indicates that the pressure equilibrium fluctuates between two metastable states, where even small perturbations can lead to different states of equilibrium in accordance with the pressure balance (Equation (1)). The aspect ratio of the keyhole stabilizes in the respective regime as the power is further shifted into the ring beam ($P_{core} = 1.0\ kW$, $P_{ring} = 2.5\ kW$). It is confirmed that the maximum bulging occurs in the center to upper region (VII). Especially for very high ring powers, there are fluctuations in the upper region of the vapor keyhole (VIII, $P_{core} = 0$, $P_{ring} = 3.5\ kW$). In general, it can be observed that the angle of inclination of the keyhole front as well as the keyhole depth decreases with increasing ring power.

Therefore, Fig. 8b shows the keyhole depth depending on core and ring power. The keyhole depth linearly decreases up to a ring power of $P_{ring} = 1.0\ kW$ following the reduction of the maximum power in the core beam. Reaching a ring power $P_{ring} = 1.5\ kW$, a significant drop in keyhole depth occurs with increased deviations, as indicated by the broadening of the 25 ∼ 75 % percentile and the lengthened whiskers, i.e., the transition mode with a metastable keyhole is reached. A further increase in ring power leads to a stable keyhole regime again with small deviations in penetration depth ($P_{core} = 0.5\cdots1.5\ kW$, $P_{ring} = 2.0\cdots3.0\ kW$). The lowest keyhole depth is reached for intensity with full ring power due to the lowest intensity reached. It should be noted that the deviations in keyhole depth as shown by the 25 ∼ 75 % percentile, whiskers and outliers, appear to be quite symmetrical, i.e., sudden changes in keyhole depth occur not only for spiking but also for a reduced penetration depth.

3.4. Absorption of laser beam power for selected power distributions

The use of a simplified 2D raytracing algorithm (Section 2.2) provides further insight into the behavior of the keyhole based on the calculation of the absorbed intensity. In order to make a general statement about the keyhole regimes, based on the high dynamics of the different power distributions considered, the median of the keyhole was

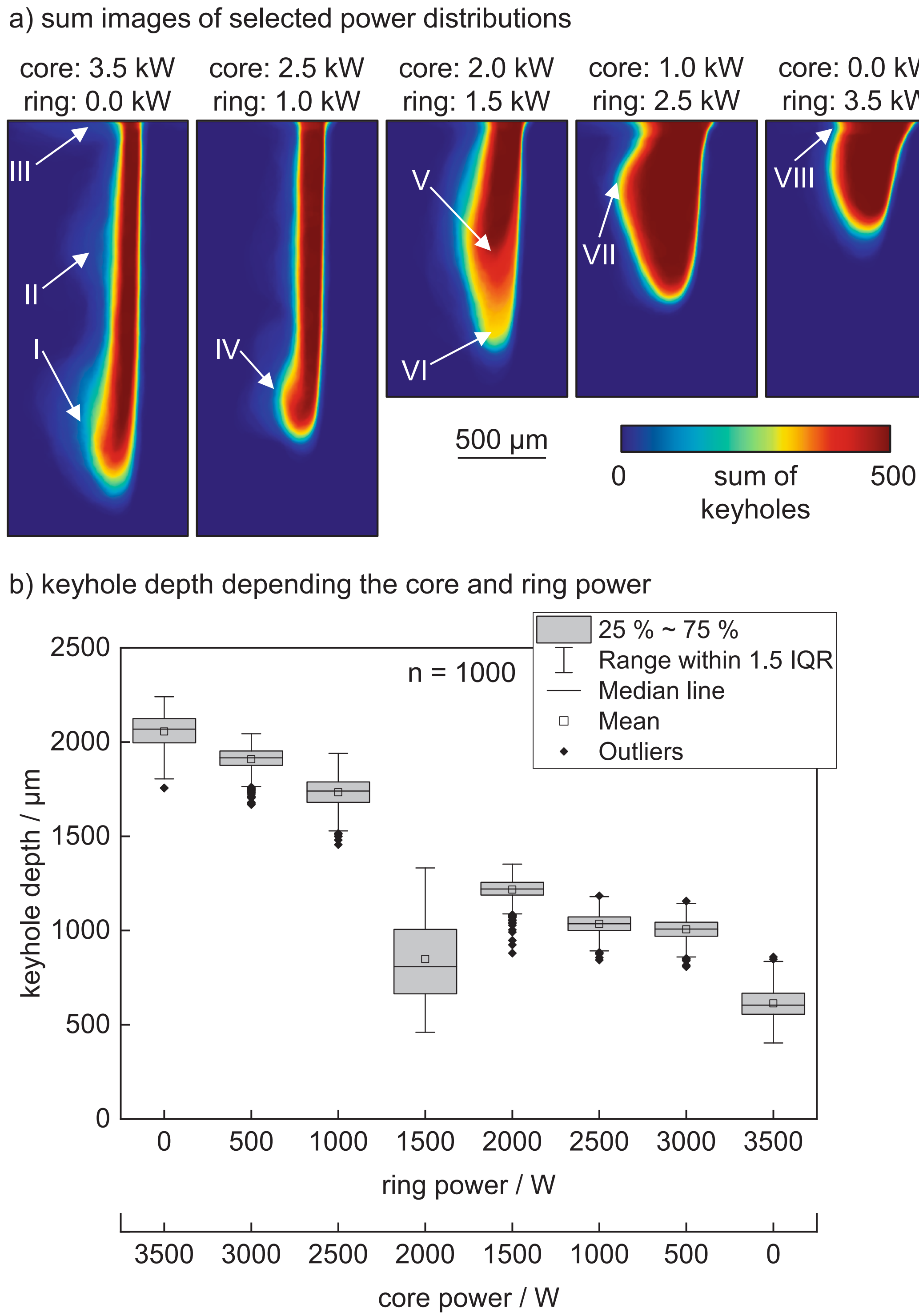

Fig. 8. a) false-color representation of sum images for selected power distributions, b) keyhole depth depending on the core and ring power.

calculated for pure core power ($P_{core} = 3.5\ kW$, $P_{ring} = 0 kW$), combined core and ring power distribution ($P_{core} = 1.5\ kW$, $P_{ring} = 2.0\ kW$) and pure ring power ($P_{core} = 0$, $P_{ring} = 3.5\ kW$). The core-ring distribution was selected intentionally to exceed the transition regime with regard to the ring power.

Fig. 9 provides a comparison between the different keyhole regimes. The pure core intensity shows the occurrence of the maximum intensity in the bottom region of the keyhole (I), with a highest absorbed intensity $\overline{I_{abs,max}}$ of $6.97 \cdot 10^6 \frac{W}{cm^2}$. The occurrence of spiking, i.e., a rapid increase in keyhole depth, which could already be recognized in the time series of Section 3.2, can be attributed to this intensity maxima at the keyhole bottom. Conversely, only a minor intensity below the evaporation threshold is absorbed in the region of the aperture on the top side. This indicates that the metal vapor escapes almost vertically and at very high velocities due to the comparatively small cross section of the keyhole. When the power is shifted into the ring beam partially ($P_{core} = 1.5\ kW$, $P_{ring} = 2.0\ kW$), the maximum absorbed intensity is reduced ($\overline{I_{abs,max}} = 5.53 \cdot 10^6 \frac{W}{cm^2}$), but remains at the keyhole bottom and is mainly affected

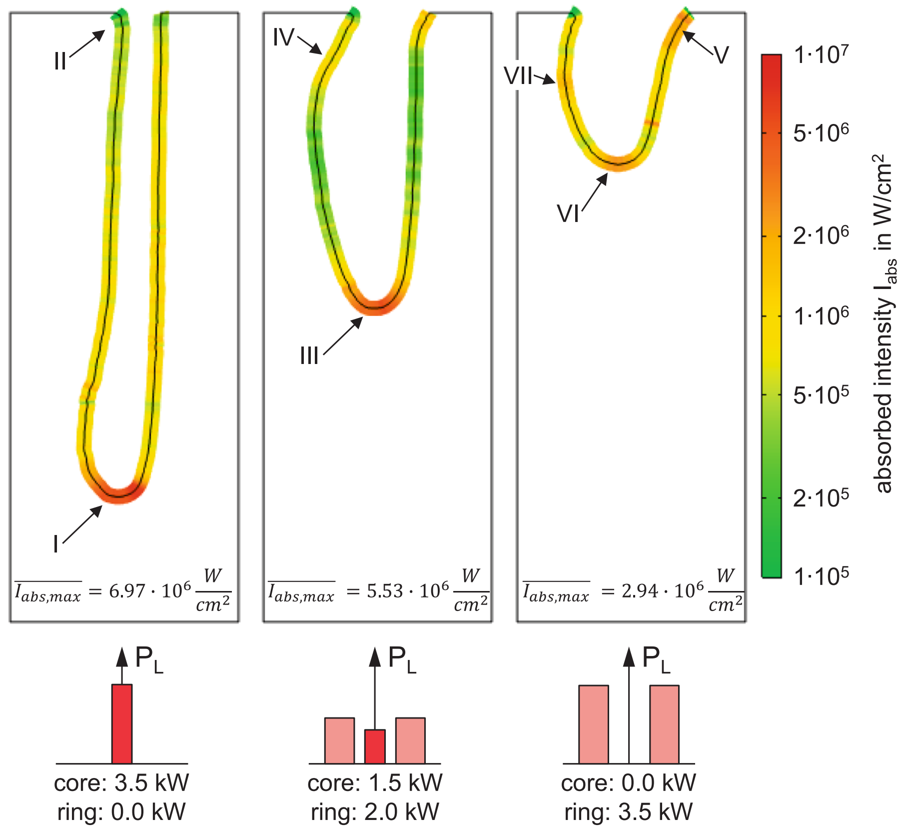

Fig. 9. Absorbed intensity in median keyhole geometries for different power distributions between core and ring based on a simplified 2D ray tracing model.

by the core beam (III). Higher absorbed intensities are detected in particular at the front of the aperture and in the upper rear area due to reflections within the keyhole (IV). The high intensity of approx. $1 \cdot 10^6 \frac{W}{cm^2}$ indicates that additional evaporation processes may support the keyhole opening pressures and could positively affect fluctuations.

A further shift of power into the ring beam is accompanied by a corresponding shift in the absorbed intensity. Firstly, the maximum absorbed intensity is located at the aperture of the keyhole in the welding direction (V). This has already been assumed in Section 3.1 in connection with the momentum transfer between the metal vapor escaping perpendicularly from the keyhole front and the melt pool wave at the rear wall. $\overline{I_{abs,max}}$ ($2.94 \cdot 10^6 \frac{W}{cm^2}$) is reduced compared to the core-dominated regimes due to the increased cross-section of the ring compared to the core beam. Secondly, the intensity absorbed at the keyhole bottom is reduced as well. The local maximum at position VI can be attributed to the backward ring beam and the elongated keyhole front. Thirdly, the local maximum at the middle of the keyhole rear wall is again caused by reflections within the keyhole (VII). The pure ring intensity shows a more uniform absorption within the keyhole, simultaneously accompanied by a reduced maximum intensity. Overall, the appearance of the keyhole resulting from the pure ring process also indicates that significantly lower pressures are required to keep the keyhole open. The larger beam diameter produces a wider melt pool [13], resulting in reduced flow velocities and, consequently, lower hydrodynamic pressure. The reduction in welding depth is accompanied by a decrease in hydrostatic pressure. The increased interaction between the metal vapor and the melt pool wave, together with the increase in the absorbed intensity at the rear wall (VII, but also for IV), leads to higher temperatures in the upper melt pool and, accordingly, to an extension of the melt pool, as observed in the numerical simulation of [13] (Video 3, discussion in the context of Section 3.1). Consequently, the increased temperatures lead to a reduction in surface tension of the copper melt [31], and thereby, a reduction of the Laplace pressure can also be assumed. At the same time, reduced surface tension can facilitate the ejection of spatter [15]. The discussion of these findings implies an effect of different power distributions on the keyhole and melt pool dynamics. Thus, the relationship between keyhole and melt pool dynamics under the three power distributions considered, will be discussed in more detail based on the resulting oscillation frequencies of the keyhole.

3.5. Oscillation frequencies of different core-ring power distributions

The behavior of the keyhole depth and aperture is described in more detail below in order to further describe the effect of the power distribution on the process dynamics based on 1,000 images. The first consideration (Fig. 10a) is the behavior of the keyhole depth in the time domain for the three power distributions that are already discussed in Section 3.4.

Significant fluctuations in keyhole depth are observed for core-dominated processes ($P_{core} = 3.5\ kW$, $P_{ring} = 0$), often including a rapid reduction of several hundred micrometers (e.g., I). On the other hand, rapid increases in weld depth are frequently observed (e.g., II), indicating spiking, as shown by examples in Section 3.2. A stabilized keyhole depth can be achieved with increasing ring power ($P_{core} = 1.5\ kW$, $P_{ring} = 2.0\ kW$), in which the fluctuations towards higher keyhole depths seem to be particularly reduced and a relatively constant penetration depth is reached. The pure ring ($P_{core} = 0$, $P_{ring} = 3.5\ kW$) process behave differently, i.e., there are phases of relatively uniform keyhole depths (e.g., 0⋯20 ms) as well as more steady changes over larger time periods, e.g., a quasi-linear increase from 40⋯50 ms.

An analysis of the associated frequency domain (Fig. 10b), as shown by amplitude over frequency, is expected to provide further insights into process dynamics, as significant peaks are expected, to indicate high process dynamic related to defects. Thereby, significant differences

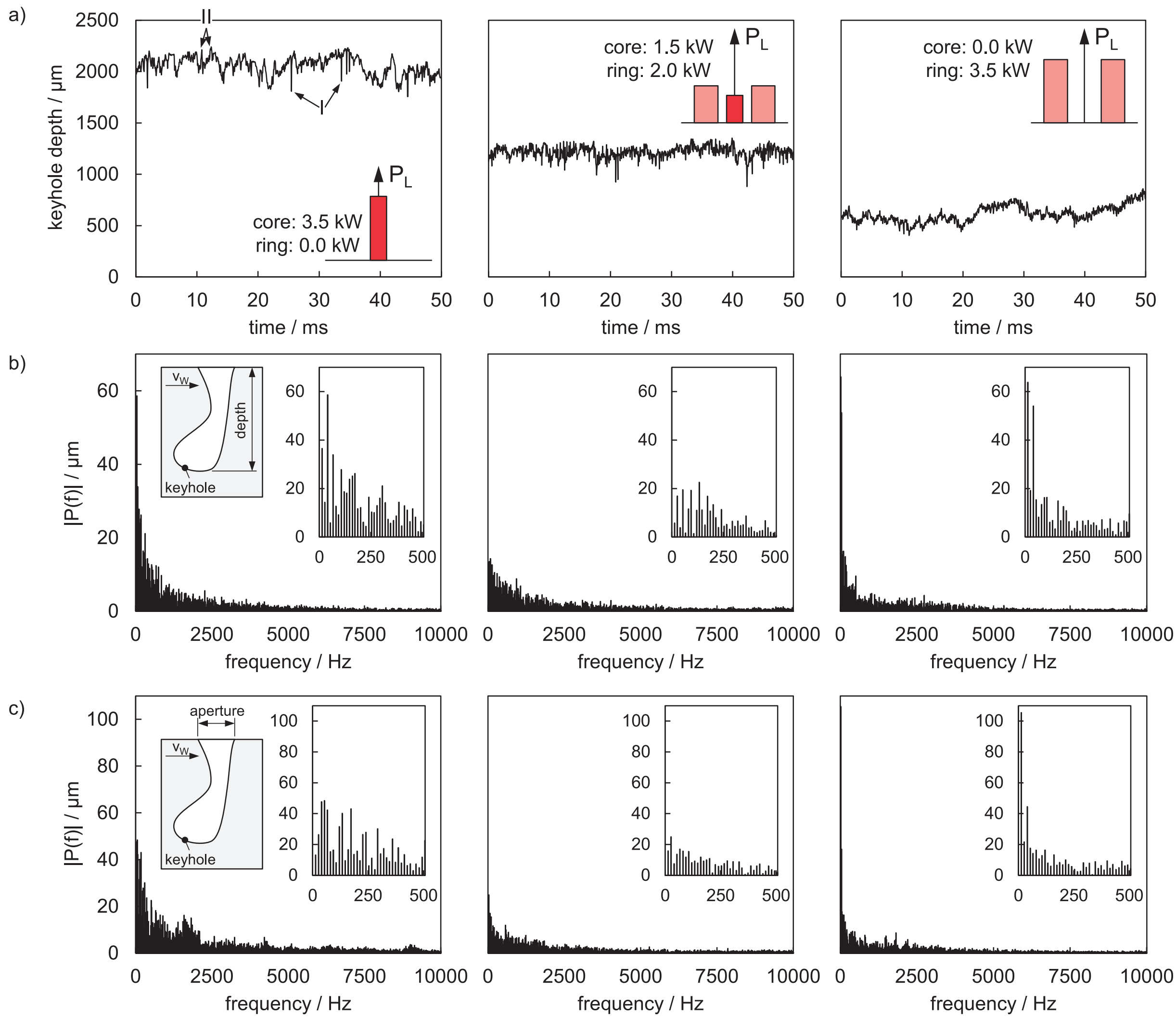

Fig. 10. a) time domain of keyhole depth, b) frequency domain of keyhole depth and c) frequency domain of keyhole aperture for different power distributions. each column represents the results of the power distribution given in a). the details in the frequency domain (b, c) show a magnified view of frequencies up to 500 Hz.

between the power distributions considered can be observed. It should be noted that at a frame rate of 20,000 images per second, maximum frequencies of up to 10,000 Hz can be determined.

The pure core process ($P_{core} = 3.5\ kW$, $P_{ring} = 0$), reveals high amplitudes in the low frequency range, which decrease exponentially with increasing frequency. The dominant frequency was identified at 40 Hz at an amplitude of approx. 60 µm. The pure ring process ($P_{core} = 0$, $P_{ring} = 3.5\ kW$) also shows high amplitudes for relatively low frequencies, with dominant peaks at approx. 13 and 40 Hz. However, the remaining frequency band shows reduced amplitudes compared to the pure core process. This illustrates the stronger tendency for more gradual changes in keyhole depth compared to the more abrupt changes that occur more frequently in the core-dominated processes. In contrast, the core-ring power distribution ($P_{core} = 1.5\ kW$, $P_{ring} = 2.0\ kW$) shows significantly reduced amplitudes, consistent with the expectation from the time domain. In general, the amplitudes are less distinct at higher frequencies.

Fig. 10c represents the corresponding results for the keyhole aperture. A quasi-exponential course of the amplitude over the frequency is again noticeable for the pure core process ($P_{core} = 3.5\ kW$, $P_{ring} = 0$). No distinct dominant frequency can be identified, as different frequencies show comparably high amplitudes – an indication of stronger fluctuations in this process regime due to varied absorption conditions and upward-oriented movement of bulges, as explained in Section 3.3. In comparison, the pure ring process ($P_{core} = 0$, $P_{ring} = 3.5\ kW$) shows a very clear dominant frequency at about 13 Hz with an amplitude of more than 100 µm. This significantly higher amplitude can be explained by increased fluctuations of the enlarged aperture due to the interaction of the escaping metal vapor with the rear melt pool wave, which is consistent with the considerations in Sections 3.1 and 3.4. In contrast, the core-ring power distribution ($P_{core} = 1.5\ kW$, $P_{ring} = 2.0\ kW$) again reveals significantly reduced amplitudes – an indication of a more stable process with reduced fluctuations. The changed absorption conditions in the keyhole (lower maximum intensity at the bottom of the keyhole, stabilization of the aperture) with less interaction of the metal vapor with the rear melt pool wave together with a reduced Laplace pressure (Section 3.4) can explain this behavior.

4. Model concept

The findings on the phenomena relevant to the effects of concentric power distributions in laser beam welding of copper can be summarized in a model concept (Fig. 11). This model concept considers three cases: a core-dominated process, a ring-dominated process, and a stabilized core-ring process characterized by significantly reduced fluctuations regarding depth and aperture. This model concept provides a comprehensive overview of the relevant effects even transferable to industrial applications.

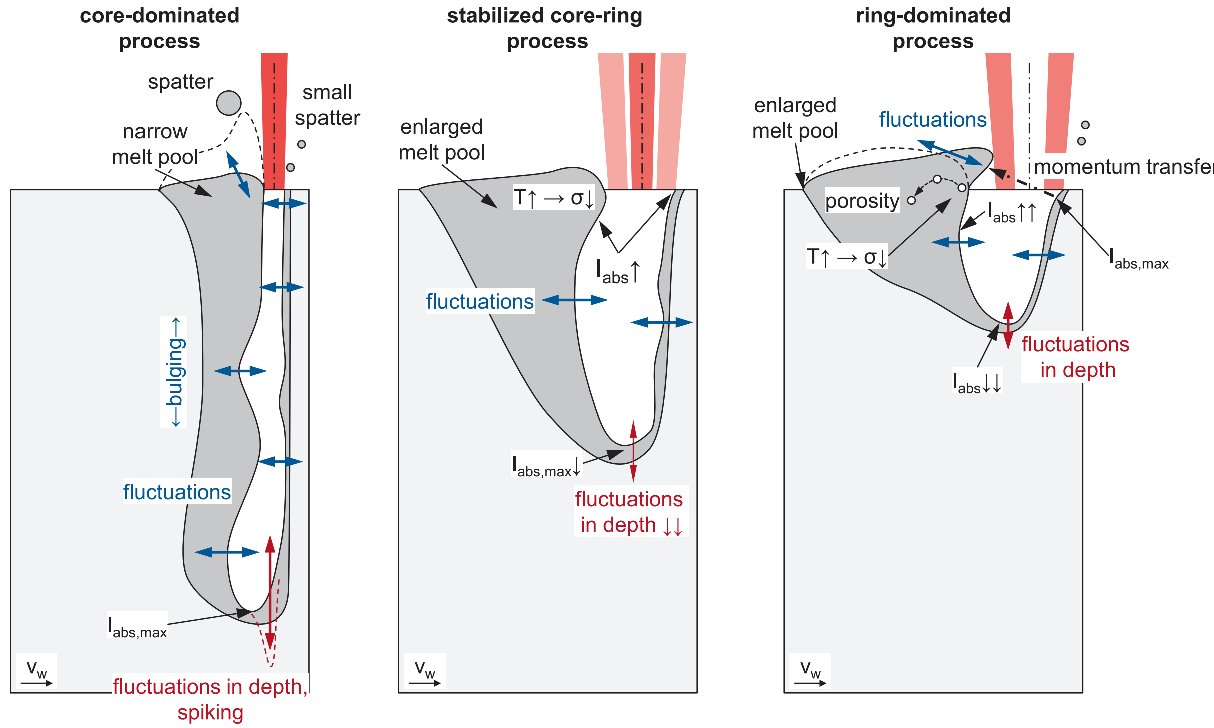

Fig. 11. Model concept for the effect of concentric power distributions on keyhole and process dynamics for three exemplary scenarios: core-dominated process, stabilized core-ring process, ring-dominated process.

- Core-dominated process: The core-dominated process is characterized by strong fluctuations of the keyhole, which cause sudden changes in absorption conditions and result in bulging. It should be noted that these fluctuations also occur at the front of the keyhole. Spatter and melt ejections can occur when bulges reach the surface, i.e., fluctuations at the keyhole aperture are accompanied by increased melt pool movement (indicated by the dashed line). The high velocity of the metal vapor within the narrow keyhole width also causes small spatters to be detached from the keyhole walls. The intensity of the laser beam is largely absorbed at the bottom of the keyhole, which can lead to spiking.
- Stabilized core-ring process: The stabilized core-ring process is characterized by a significant change in absorption within the keyhole. The maximum absorbed intensity is reduced, which explains less depth variation and a decrease in penetration depth, contributing to lower hydrostatic pressure. The leading edge of the keyhole in welding direction shows significantly reduced fluctuations due to the increased absorption in this region. In addition, the upper keyhole rear wall fluctuates less because of increased absorption in this region. This can be expected to increase the melt pool temperature and consequently reduce the Laplace pressure. The changes result in an elongated and wider pool without large melt pool build-ups. Hence, the metal vapor escaping from the leading edge does not interact strongly with these rear waves and does not add additional process fluctuations. In addition, the widened melt pool increases the cross-sectional area of the melt flow, which can contribute to reduced hydrodynamic pressure. This reduces the closing pressures acting on the keyhole. It should be noted that fluctuations still occur at the front and back of the keyhole, but their negative effect on imperfections is limited by the stabilized aperture.
- Ring-dominated process: The ring-dominated process is accompanied by a further change in the aspect ratio of the keyhole, i.e. the depth decreases, and the aperture increases. This also changes the volume ratio of the keyhole to the melt pool, accompanied by an increased displacement of melt, and resulting in a larger melt pool build-up (dashed line). The modified absorption conditions and the adapted pressure balance result in this keyhole shape: First, by reducing the maximum intensity at the bottom of the keyhole, which results in a more gradual variation of the welding depth. Second, there is an increase in absorption in the rear keyhole area, which also results in an increased melt pool temperature. Third, the maximum absorbed intensity is shifted to the leading edge of the keyhole. The metal vapor escaping perpendicular to the leading edge induces strong fluctuations in the rear melt pool wave due to momentum transfer. At the same time, the increased weld pool temperature facilitates melt ejection and small spatter formation due to the reduced surface tension. In addition, the increased temperature and the large keyhole aperture can promote nitrogen dissociation leading to pore formation in Cu-ETP (Section 3.1). Pores form in the upper keyhole region or in the waves of the melt pool facing the keyhole in this region. The pores are then transported backwards by the melt pool flow (indicated by dotted line).

5. Conclusion

This paper focuses on the effect of concentric power distributions consisting of core and ring beams on the keyhole behavior during laser beam welding of copper. High-speed synchrotron X-ray imaging provided a high spatial and temporal resolution for the analysis of phenomena hidden within the workpiece, thus providing deep insights into understanding the process dynamics and their relationship with the power distribution.

The effects of concentric power distributions were systematically studied by adjusting the power between core and ring beam in steps of $0.5kW$ while maintaining the total laser beam power constant at $3.5kW$. The evaluation of the images was carried out based on the radiography to describe fundamental aspects qualitatively. On the other hand, the extraction of keyhole geometries by pixel classification and subsequent image processing allowed for a time-dependent evaluation and the quantification of effects by extracting the keyhole geometry. On this basis, the keyhole geometry and the bulging behavior and keyhole depth could be described in more detail and depending on the core-ring power distribution. Furthermore, the calculation of the median keyhole geometry for selected power distributions enabled the calculation of the spatially absorbed intensity within the keyhole by means of raytracing, providing further insights for describing the behavior. A model concept was developed to summarize the results in a generally valid way.

In general, the core-dominated processes showed a highly dynamic keyhole behavior characterized by strong fluctuations, bulging primarily in the lower keyhole region, and maximum absorbed intensities at the keyhole bottom. Conversely, ring-dominated processes revealed reduced keyhole depths, with the maximum intensity shifted to the keyhole aperture. While this process reduced depth fluctuations, the interaction between metal vapor and the melt pool wave at the keyhole rear wall increased. However, in combining core and ring beam, a stabilized core-ring process that effectively balances the opening and closing pressures acting on the keyhole can be reached. This configuration achieved substantial reductions in both keyhole depth and aperture oscillations by adjusting the absorbed intensity at the keyhole bottom and along the radius at the keyhole aperture in welding direction, as well as melt pool geometry and temperature distribution. These findings were summarized in a model concept to provide a comprehensive explanation of the different phenomena acting for describing the effect of concentric intensity distributions.

Data availability statement

The data used in this study are available upon reasonable request from the corresponding author.

CRediT authorship contribution statement

Klaus Schricker: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Leander Schmidt:** Software, Investigation, Formal analysis. **Falk Nagel:** Methodology, Investigation, Conceptualization. **Christian Diegel:** Investigation. **Hannes Friedmann:** Investigation. **Marc Seibold:** Software, Investigation. **Peter Hellwig:** Investigation. **Fabian Fröhlich:** Investigation. **Peter Kallage:** Resources, Investigation. **Yunhui Chen:** Investigation. **Herwig Requardt:** Investigation. **Alexander Rack:** Writing – review & editing, Resources, Investigation. **Jean Pierre Bergmann:** Writing – review & editing, Resources, Conceptualization.

Author declarations

Ethics approval

Not applicable.

Funding Source Declaration

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Permission note

Not applicable.

Author contributions

Conceptualization: K.S., F.N. and J.P.B.; Data curation: K.S.; Formal analysis: K.S. and L.S.; Investigation: K.S., L.S., C.D., H.F., M.S., P.H., F. F., F.N., P.K., A.R., H.R. and Y.C.; Methodology: K.S. and F.N.; Resources: A.R., P.K. and J.P.B.; Software: K.S., L.S. and M.S.; Visualization: K.S.; Writing – original draft: K.S.; Writing – review & editing: A.R. and J.P.B.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.optlastec.2025.113999.

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